Supplementary Information

BIOREDIS: rapid and customizable design of nucleic acid therapeutics

Supplementary Tables

Codons optimisation

tool	gene	GC	freq	changed_bases %	changed_codons %	MFE [kcal/mol]
Native	KIT	44.9588	0.38	0	0.0	-777.3
Native	PAX3	57.6733	0.38	0	0.0	-383.3
BIOREDIS	KIT	59.2936	0.46	24	62.0	-1015.3
BIOREDIS	PAX3	57.9208	0.41	23	56.9	-403.8
CodonTransformer	KIT	55.4184	0.44	21	56.0	-1006.1
CodonTransformer	PAX3	61.4686	0.41	20	51.0	-452.2
GenScript	KIT	54.8354	0.44	23	60.0	-893.3
GenScript	PAX3	58.2508	0.41	22	54.0	-362.4
VectorBuilder	KIT	54.8011	0.44	23	59.0	-922.8
VectorBuilder	PAX3	58.9109	0.41	22	55.0	-396.7

Table S1: Data including statistics of the native sequences (*KIT, PAX3*) and optimized sequences generated using various tools.

source	RNAi_seq	specificity	GC%	score
BIOREDIS	ATAACAGCCTAATCTCGTCGC	1	47.62	10
BIOREDIS	TAACAGCCTAATCTCGTCGCC	1	52.38	8
BIOREDIS	TATTCTCATTCGTTTCATCCA	1	33.33	7
BIOREDIS	TGTAGAACTTAGAATCGACCG	1	42.86	7
BIOREDIS	TCATAAGGAAGTTGTGTTGGG	1	42.86	7
BIOREDIS	ATATTGGGCAGTGAGGAAGGG	1	52.38	7
BIOREDIS	ATATACAAGTGCTGGCTAGGG	1	47.62	7
BIOREDIS	GAAAGACAGGATTGCAGTGGG	1	52.38	7
BIOREDIS	TTTGTAGGTCAGAATCATCAC	1	38.10	6
BIOREDIS	CATTCTGCTTATTCTCATTCG	1	38.10	6
Eurofins Genomics	TTCTGCTTATTCTCATTCG	1	36.84	6
Eurofins Genomics	TATATGGCTCTTAATCTAC	1	31.58	1
Eurofins Genomics	TCTATAACTAACAACCACT	1	31.58	1

source	RNAi_seq	specificity	GC%	score
Eurofins Genomics	TTATCTACTACTTCCAAGG	1	36.84	6
Eurofins Genomics	TTAATTAAGCCATAAGCAG	1	31.58	4
Eurofins Genomics	TTCTGCATGATCTTCCTGC	1	47.37	4
Eurofins Genomics	TCTTGTATCATCATACTAC	1	31.58	-1
Eurofins Genomics	TGTATGAACACGGACAATT	1	36.84	-2
Eurofins Genomics	TATAGATCCACTGCTGGTG	1	47.37	4
Eurofins Genomics	TTAATCTCGTTAGATGAAG	1	31.58	1
GenScript	ACTATCGCAGGAAGACTCCTT	0	47.62	0
GenScript	TGGTAGGCGCGTTTCACACTT	1	52.38	-2
GenScript	TGTATACAGAAACGAAACCTT	1	33.33	1
GenScript	AGAACTCTGAACCACTAGCTT	1	42.86	1
GenScript	GTCTAAATAACAGAGTGCCTT	1	38.10	-2
GenScript	GAAACGAAACCTTATGGTGTT	1	38.10	2
GenScript	AGTCATGGCCGCATCTGACTT	1	52.38	0
GenScript	TTCCAAGGTTGTTGTGACATT	1	38.10	-1
GenScript	CTCGTTAGATGAAGTTCACTT	1	38.10	-1
GenScript	GCACGTGTATTTGCCGGTGTT	1	52.38	-1
Invivogen	AGTACTATCGCAGGAAGACTC	0	47.62	2
Invivogen	ATAACAGCCTAATCTCGTCGC	1	47.62	10
Invivogen	TAACAGCCTAATCTCGTCGCC	1	52.38	8
Invivogen	TTATTCTCATTCGTTTCATCC	1	33.33	6
Invivogen	TTGTAGAACTTAGAATCGACC	1	38.10	3
Invivogen	TTTACAGCGACAGTCATGGCC	1	52.38	7
Invivogen	ATCTCTTTCTATGTATGAGCC	1	38.10	6
Invivogen	TATTTGTAGGTCAGAATCATC	1	33.33	2
Invivogen	TTTGAGTTCAGACATGAGGGC	1	47.62	8
Invivogen	ATAACAGAGTGCCTTAAGTGC	1	42.86	6
Invivogen	ATTCTTGGAGGCGAGGAAAGC	1	52.38	5
Invivogen	ATAGATCCACTGCTGGTGTTC	1	47.62	2
Invivogen	TGGTAAGCTACTCTAGACAGC	1	47.62	1
VectorBuilder	ATAATTGGTCACTTCTGGGTC	1	42.86	5
VectorBuilder	ATATTTCTCCTGTAGTTTAGT	1	28.57	4
VectorBuilder	AATAAACCTCAAGTCCTTGGG	1	42.86	9
VectorBuilder	ATATTACTTTCATTCTCAGAC	1	28.57	3
VectorBuilder	TTTATTGTGCACGTCACTGTG	1	42.86	6
VectorBuilder	TTTAATCTCGTTAGATGAAGT	1	28.57	3
VectorBuilder	ATAATTGAAGTCACCGTGATG	1	38.10	2
VectorBuilder	ATTATTGGCATAACACATGAA	1	28.57	2
VectorBuilder	ATAAATGGAATTGCTTAAGCC	1	33.33	8
VectorBuilder	ATTTATCAGTGAAGGTTCTGT	1	33.33	3

Table S2: Data including statistics of the predicted RNAi of *KIT* gene using various tools.

source	RNAi_seq	specificity	GC%	score
BIOREDIS	AATTATATCCAGGTGAAGGCG	1	42.86	8
BIOREDIS	AAATTATATCCAGGTGAAGGC	1	38.10	8
BIOREDIS	TTTACTTCTCAGGATGCGGCT	1	47.62	7
BIOREDIS	ATTTACTTCTCAGGATGCGGC	1	47.62	7
BIOREDIS	AATTTACTTCTCAGGATGCGG	1	42.86	7
BIOREDIS	TTCAGAGTCAATATCAGAGCC	1	42.86	6
BIOREDIS	TTCTTCTCGCTTTCCTCTGCC	1	52.38	6
BIOREDIS	AAGTTTGGTACGAGTCTGGGC	1	52.38	6
BIOREDIS	TATCCAGGTGAAGGCGAAACG	1	52.38	6
BIOREDIS	TGAAGGCGAAACGGAAAGGCG	1	57.14	6
Eurofins Genomics	TTCTTCTCCACGTCAGGCG	1	57.89	6
Eurofins Genomics	TCTTCCTCTTCTCCACTGC	1	52.63	3
Eurofins Genomics	TGTTCTAGCTCCTCGATGA	1	47.37	2
Eurofins Genomics	TTCCTCTTCTCCACTGCTT	1	47.37	0
Eurofins Genomics	TACAACACATTGTTGTTGG	1	36.84	6
Eurofins Genomics	TATCAGAGCCTTCATCTGA	1	42.11	3
Eurofins Genomics	TGTTCTGCTGTGAAGGTGG	1	52.63	3
Eurofins Genomics	GTCAAGGATTCCTCCATTC	1	47.37	-1
Eurofins Genomics	TATGTGAATGCATGTCACT	1	36.84	4
Eurofins Genomics	CATGTGAACACTTCTGTTC	1	42.11	1
GenScript	CAGAGTCAATATCAGAGCCTT	1	42.86	-1
GenScript	ACCATGTGAAACCATTGCCTT	1	42.86	0
GenScript	TCGATCACAGACCGCGTCCTT	1	57.14	-1
GenScript	GTGCTTTGGTGTACAGTGCTT	1	47.62	-2
GenScript	GTCAGGCGTTGTCACCTGCTT	1	57.14	-2
GenScript	AGGATTCCTCCATTCTTGCTT	1	42.86	-2
GenScript	CCTTTACAGCTAGTCTAGCTT	1	42.86	-2
GenScript	ACTCGCTACGTCATAGTTCTT	1	42.86	1
GenScript	ACTCACTGACGGCACGGTGTT	1	57.14	1
GenScript	GCTGTGAAGGTGGTTCGGCTT	1	57.14	-1
Invivogen	AAGTTTGGTACGAGTCTGGGC	1	52.38	6
Invivogen	AACTAAATTCGGTACTATGTC	1	33.33	3
Invivogen	ATTTGTCTCGGATTTCCCAGC	1	47.62	6
Invivogen	TTCACTTACGCGATATCTGGC	1	47.62	6
Invivogen	TTATATCGCCTTGGGCATTGC	1	47.62	6
VectorBuilder	AATTTACTTCTCAGGATGCGG	1	42.86	7
VectorBuilder	AAATAACACTGCCTTCCACCT	1	42.86	6
VectorBuilder	TTATATCCAGGTGAAGGCGAA	1	42.86	1
VectorBuilder	TTTATTGCTCCAGGTCTTCCT	1	42.86	7
VectorBuilder	TATTTAGGAGGTCCTTTACAG	1	38.10	4
VectorBuilder	ATAAATGTCAGGGTAATGAGT	1	33.33	3
VectorBuilder	TTATTTCAATTTCCAGTGTGT	1	28.57	3
VectorBuilder	AATTATCTGATCTGGTCTCTA	1	33.33	3
VectorBuilder	TAATTTGTCTCGGATTTCCCA	1	38.10	6
VectorBuilder	TTATATCGCCTTGGGCATTGC	1	47.62	6

Table S3: Data including statistics of the predicted RNAi of *PAX3* gene using various tools.

Supplementary Figures

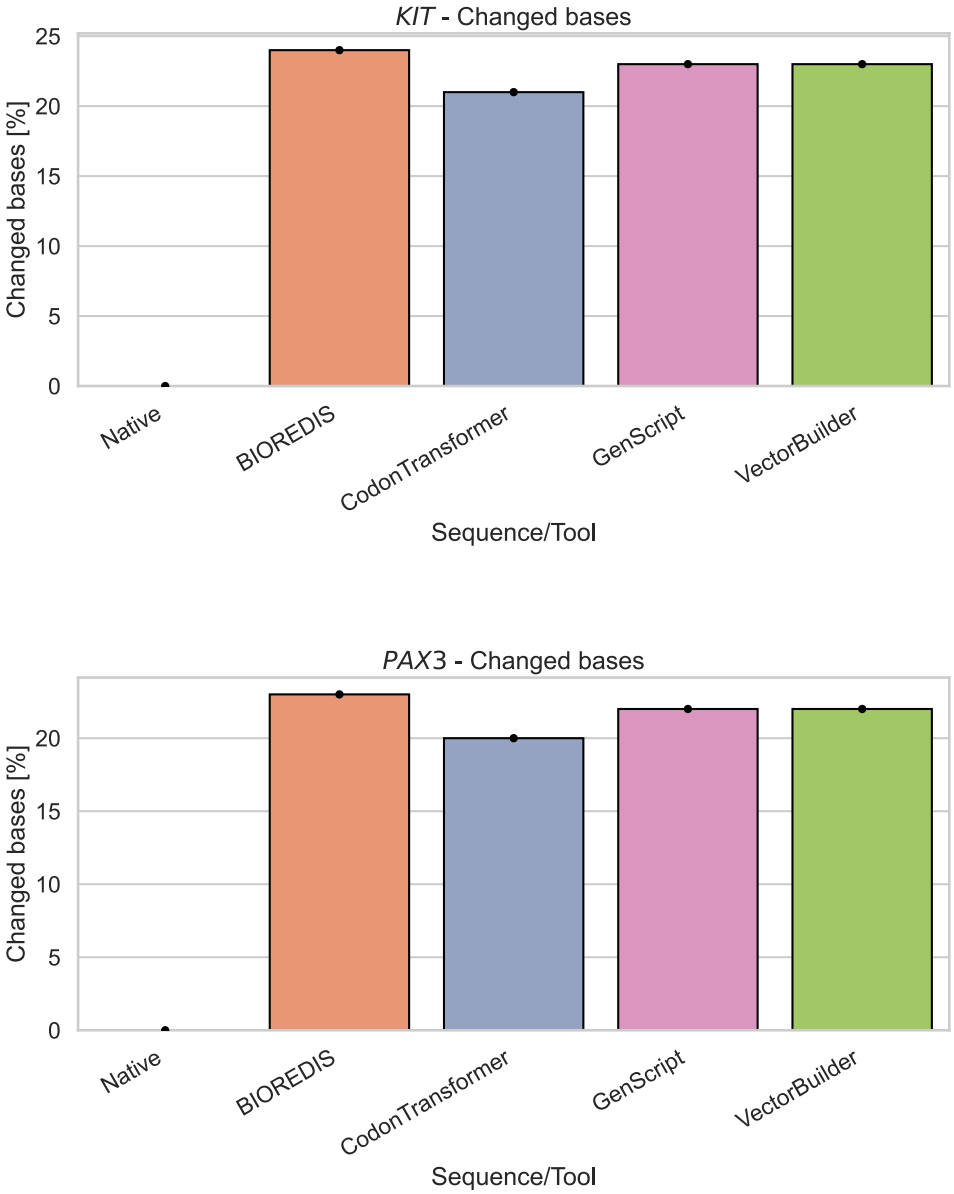

Figure S1. Percent of nucleotide changes in optimised sequences relative to the native sequence of the *KIT* and *PAX3* genes.

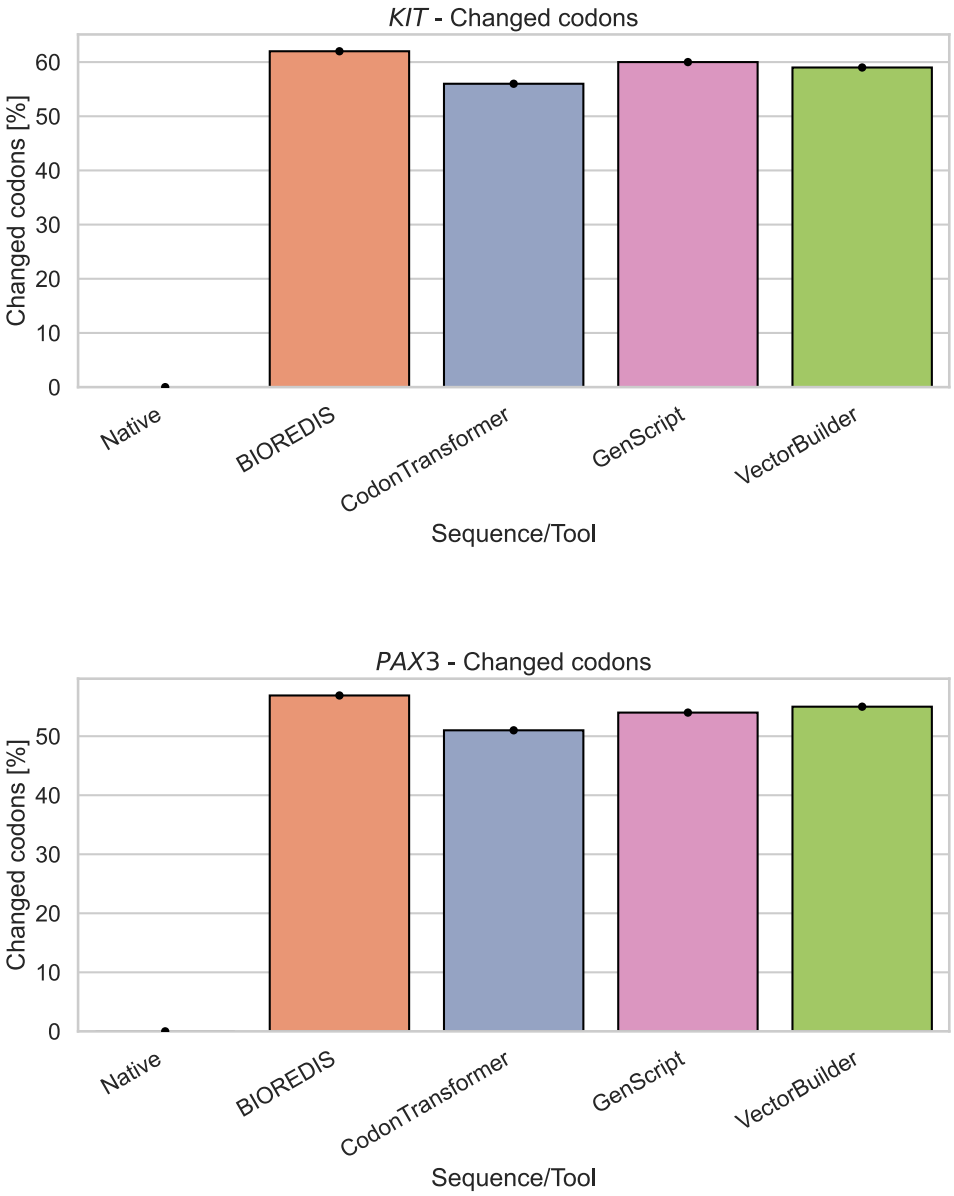

Figure S2. Percent of codon changes in optimised sequences relative to the native sequence of the *KIT* and *PAX3* genes.

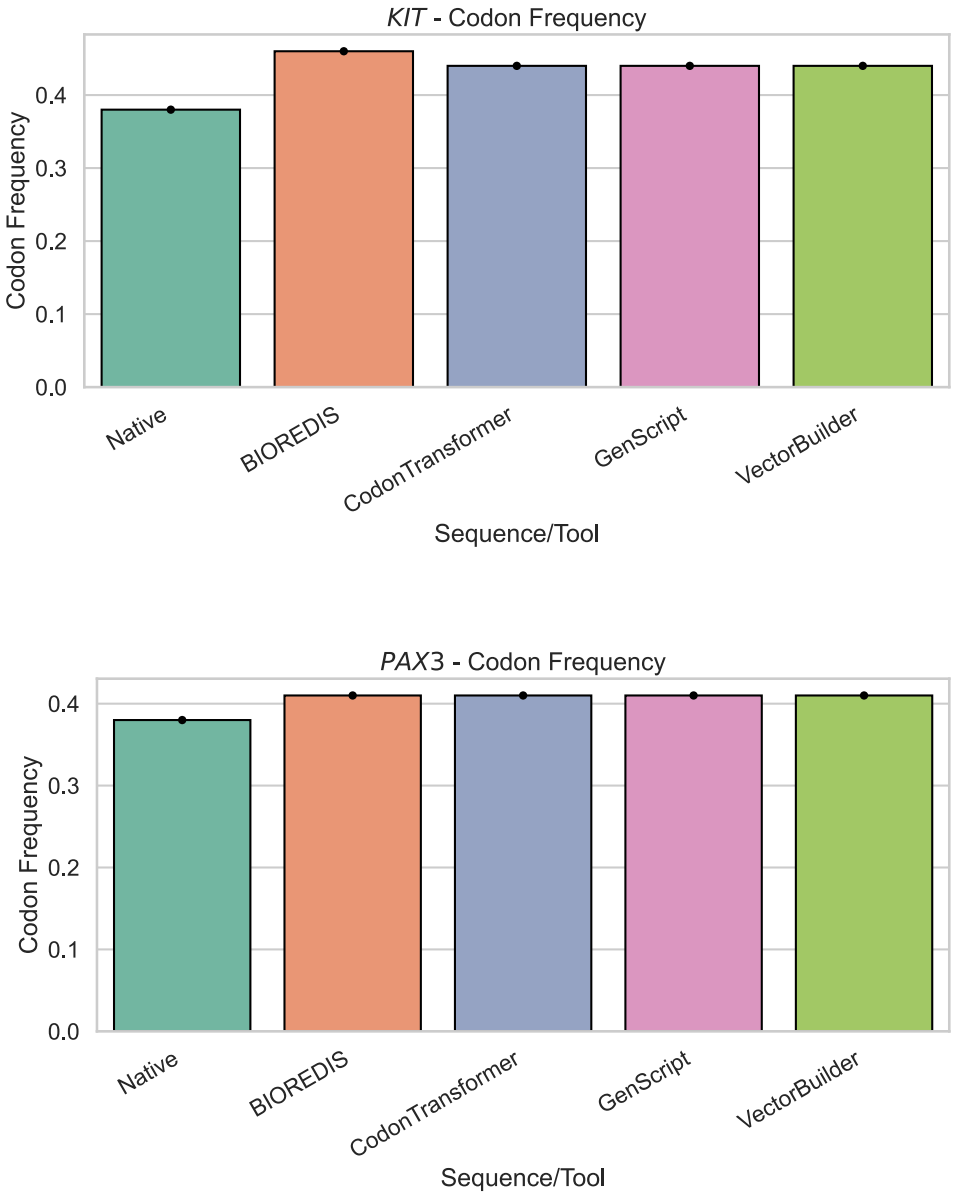

Figure S3. Codon frequency values in optimised and native sequences of the *KIT* and *PAX3* genes

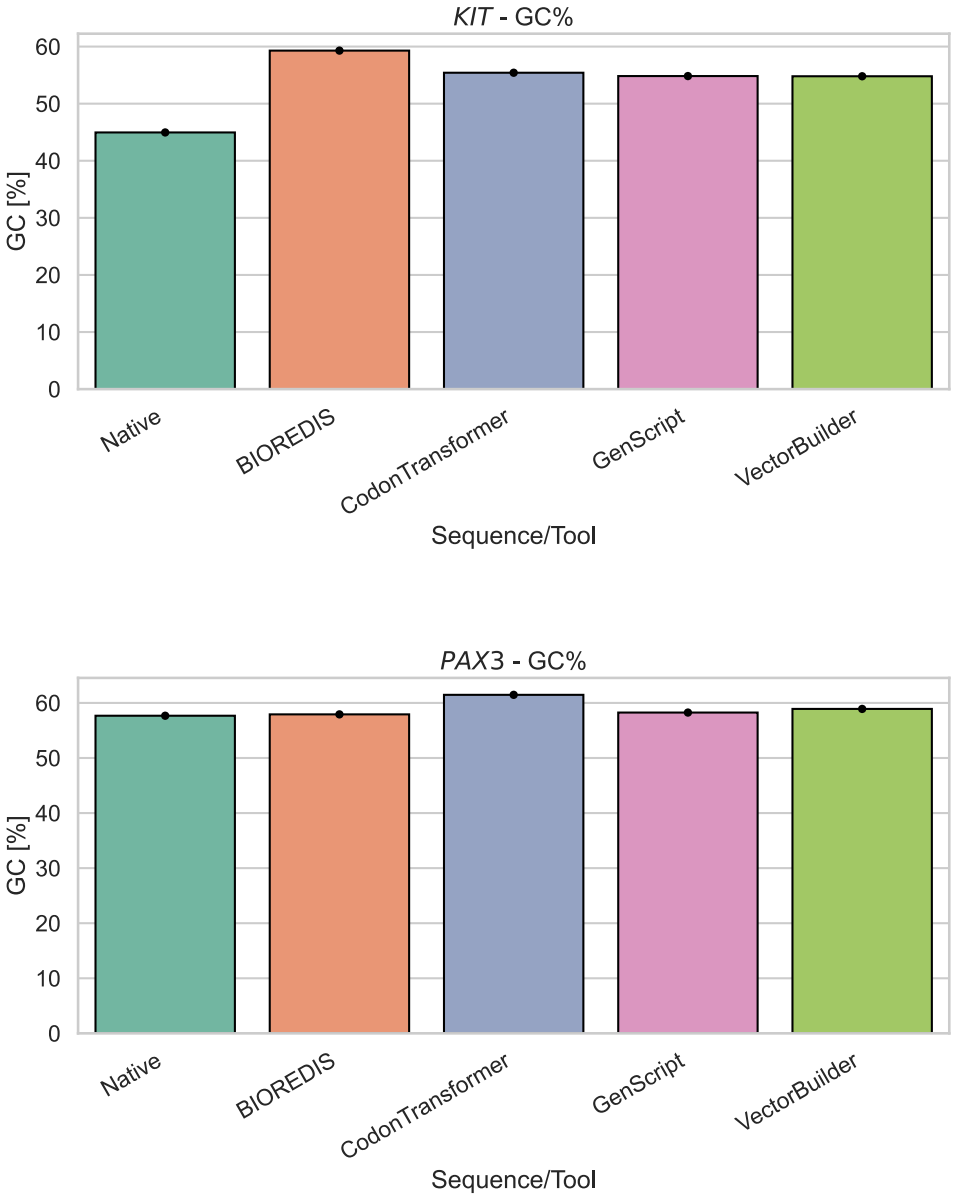

Figure S4. GC content [%] in optimised and native sequences of the *KIT* and *PAX3* genes

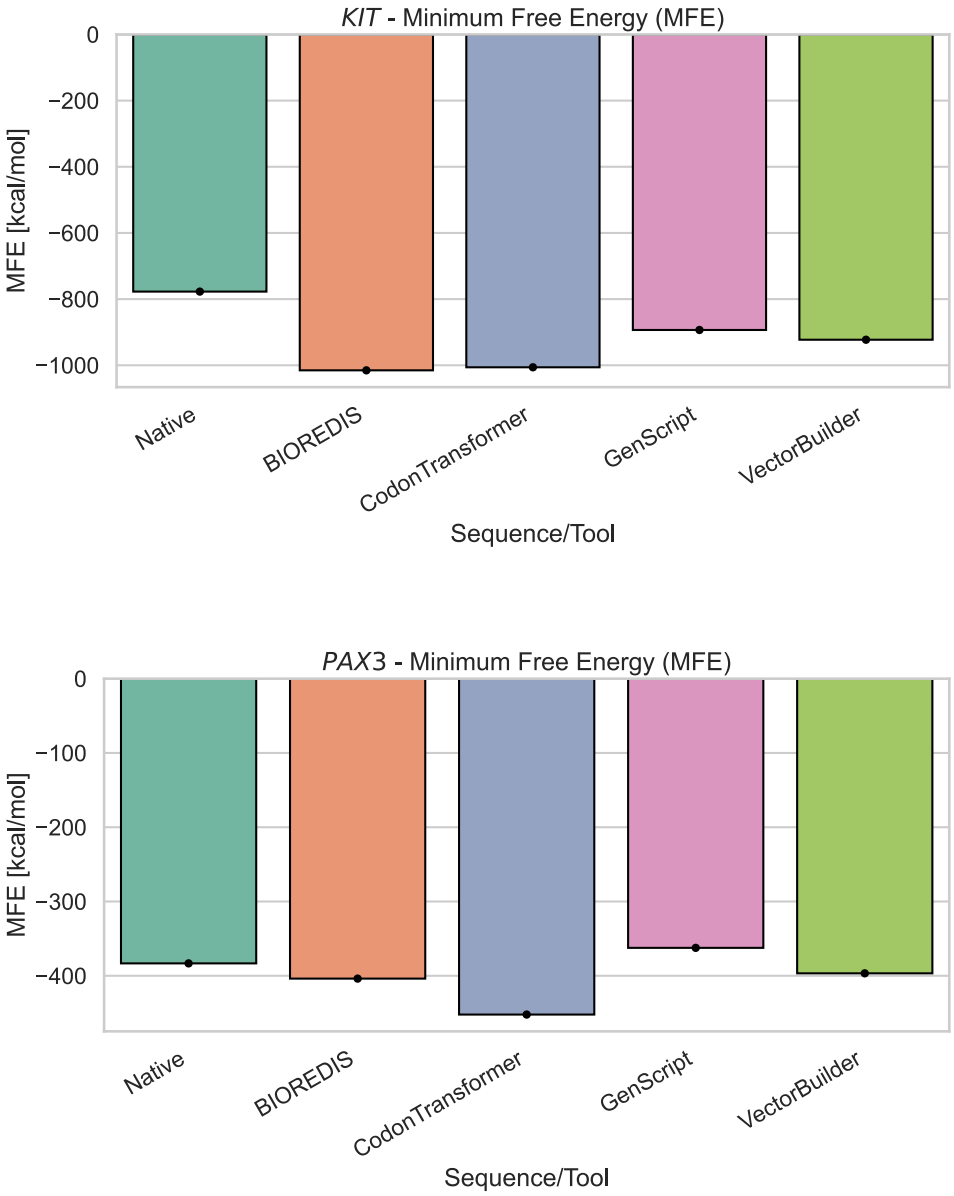

Figure S5. MFE [Minimum Free Energy] values in optimised and native sequences of the *KIT* and *PAX3* genes

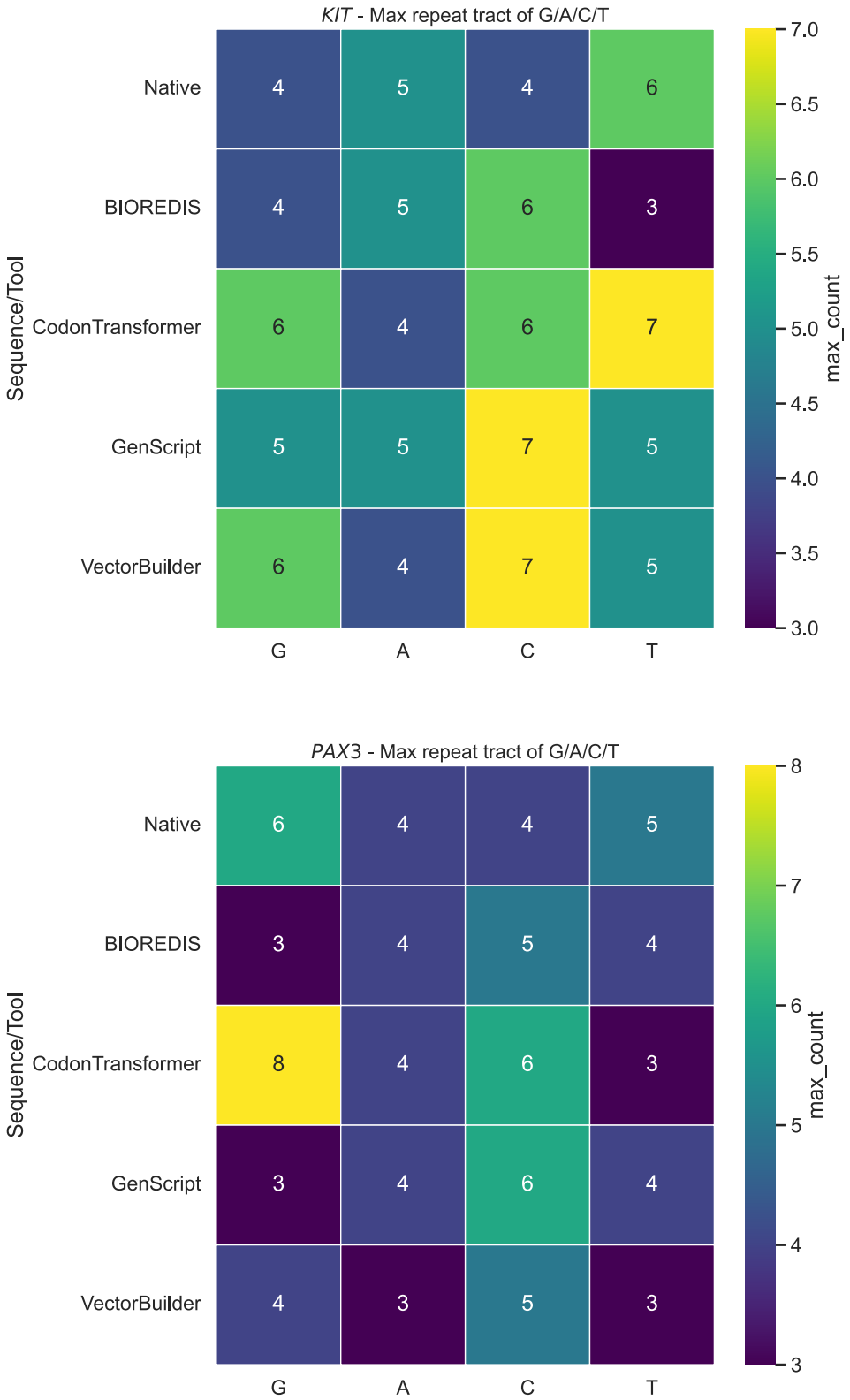

Figure S6. Heatmap of longest consecutive A/C/T/G tracts in optimised sequences of the *KIT* and *PAX3* genes.

RNAi (siRNA/shRNA) prediction

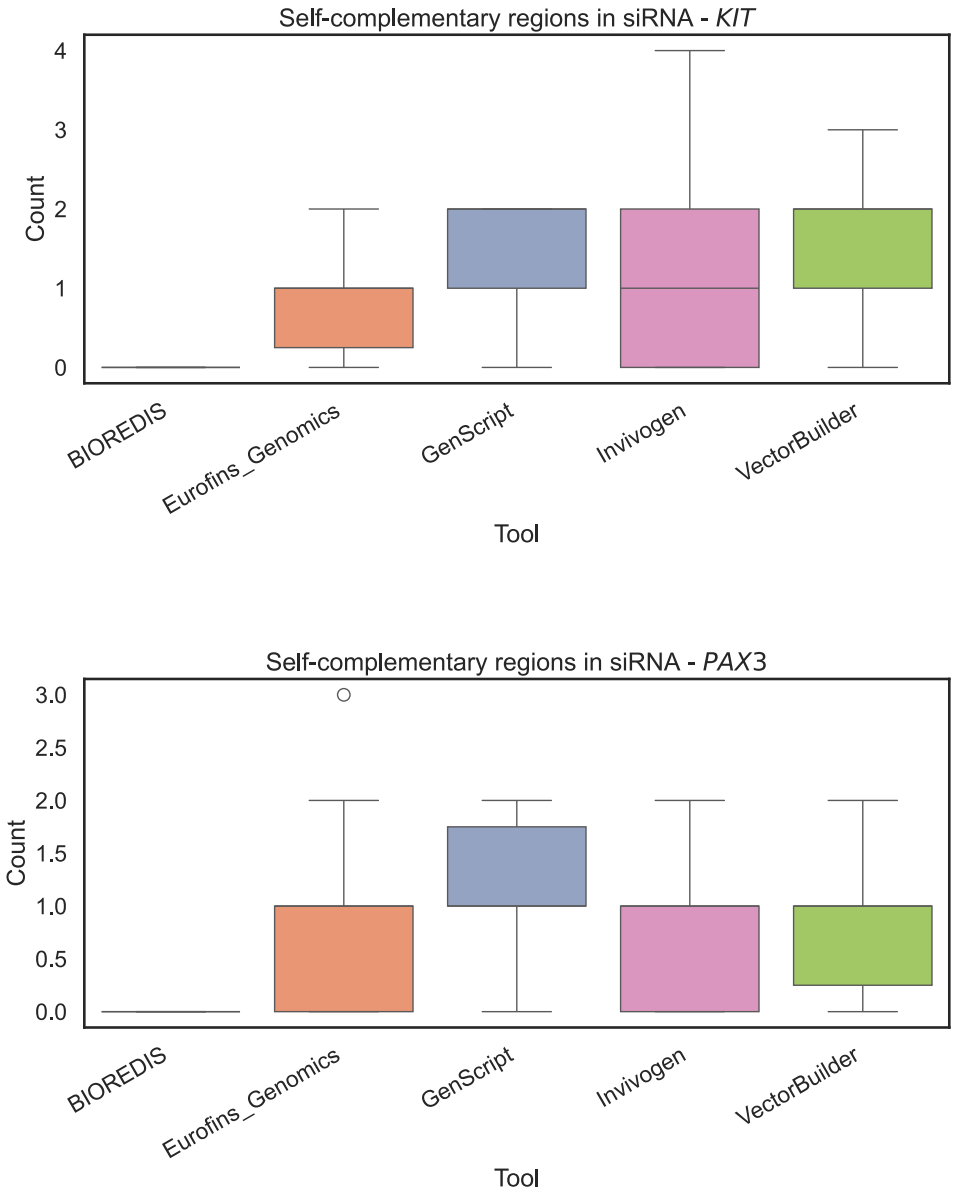

Figure S7. Counts of self-complementary regions in predicted RNAi sequences targeting the *KIT* and *PAX3* genes.

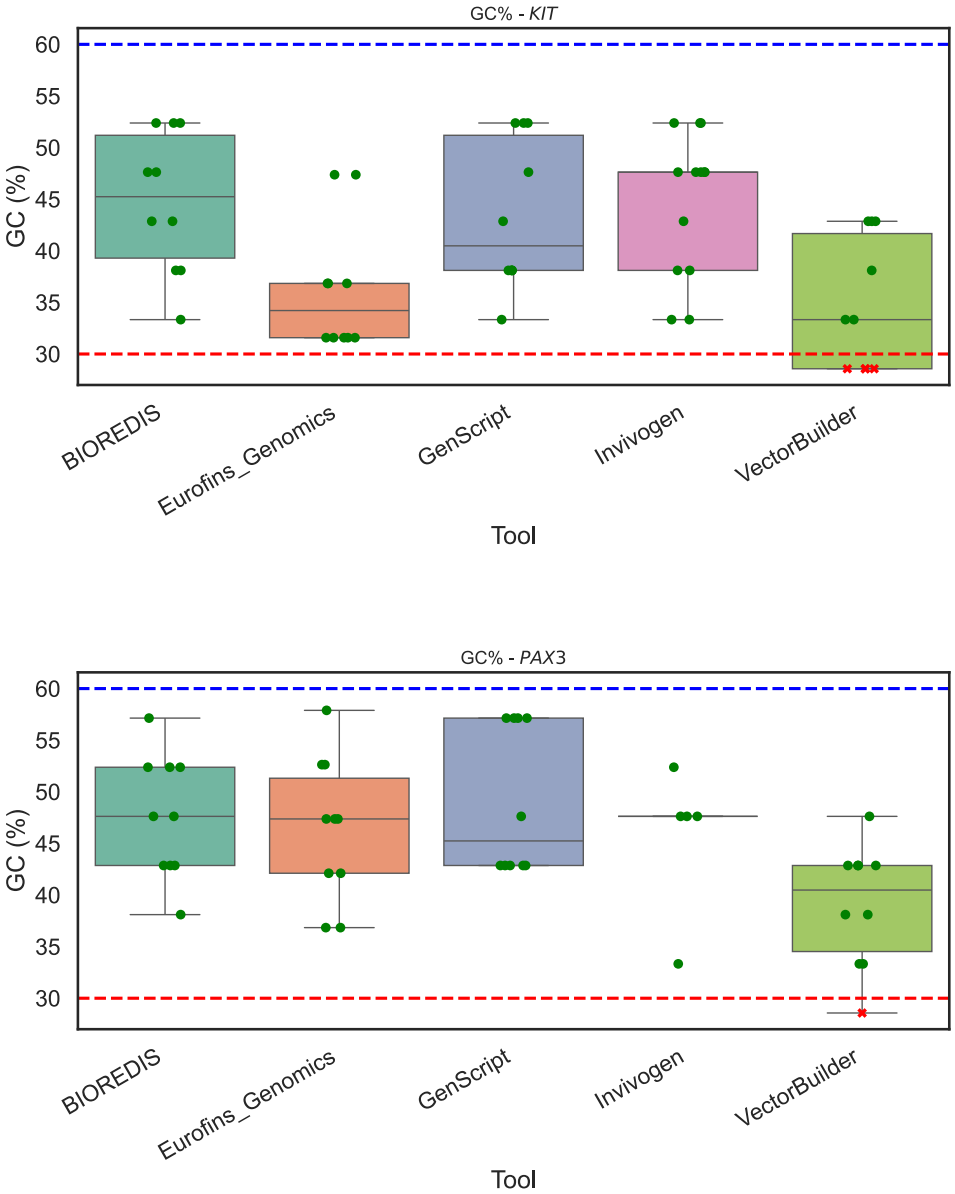

Figure S8. GC content [%] in predicted RNAi sequences targeting the *KIT* and *PAX3* genes.

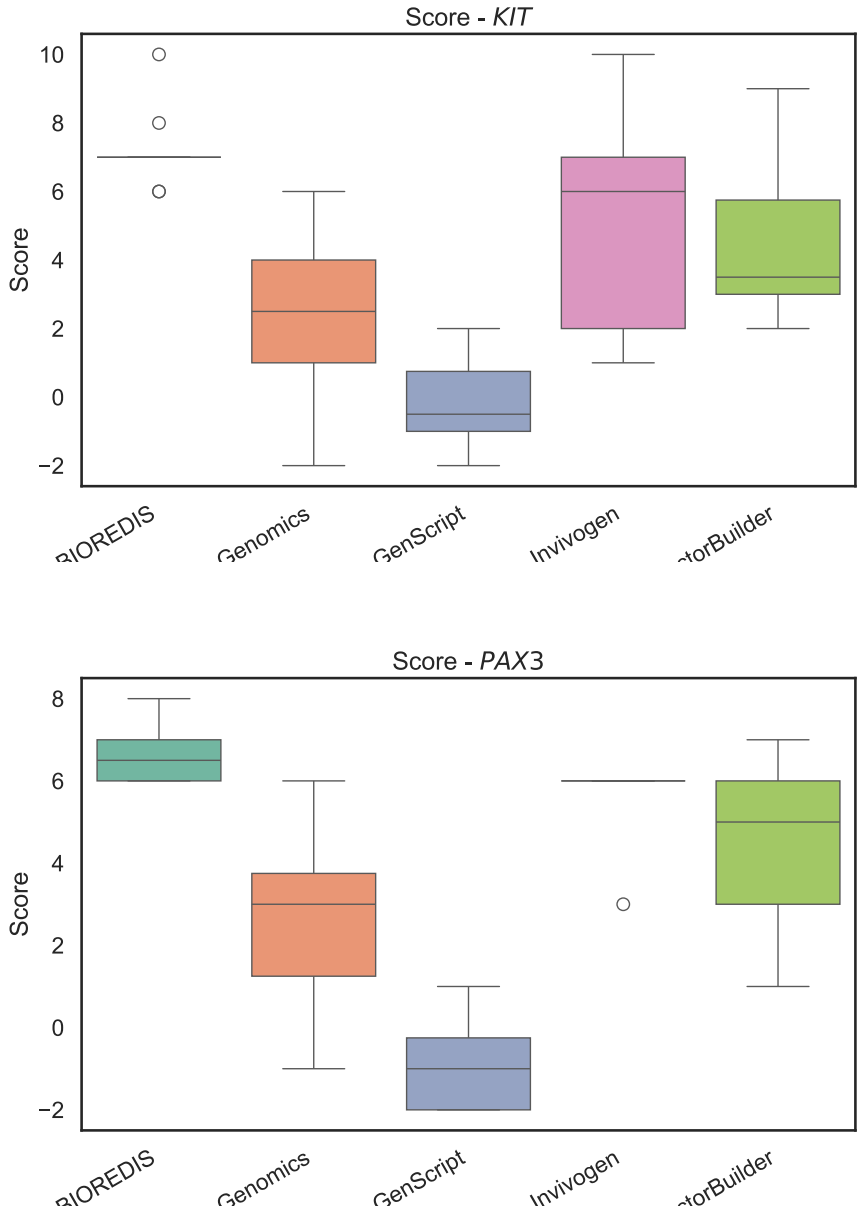

Figure S9. Nucleotide positions score in predicted RNAi sequences targeting the *KIT* and *PAX3* genes based on series of publications: DOI:10.1093/nar/gkn902; DOI:10.1093/nar/gkh247; DOI:10.1038/nbt936;; DOI:10.1016/j.bbrc.2004.02.157